作业纸

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9.15 (1) 热运动平均平动动能

$\bar{\varepsilon}_t = \frac{3}{2}kT = \frac{3}{2}\times 1.38\times10^{-23}\times 290\ J = 6.00\times10^{-21}\ J$

平均转动动能

$\bar{\varepsilon}_r = \frac{2}{2}kT = \frac{2}{2}\times 1.38\times10^{-23}\times 290\ J = 4.00\times10^{-21}\ J$

平均总动能

$\bar{\varepsilon}_k = \bar{\varepsilon}_t + \bar{\varepsilon}_r = 10.00\times10^{-21}\ J$

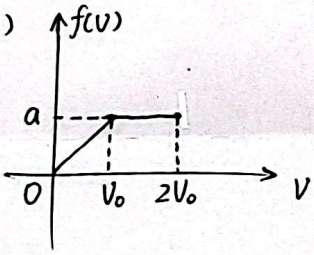
(2) 内能

$E = \frac{i}{2}\nu RT = \frac{5}{2}\times\frac{8.5}{28}\times 8.31\times 290\ J = 1.83\times10^{3}\ J$

(3) 轨道动能

$E_k = \frac{1}{2}mv^2 = \frac{1}{2}\times 8.5\times10^{-3}\times\left(\frac{65}{3.6}\right)^2 J = 1.39\ J$

9.18 (1)



由归一化条件有

$\frac{1}{2}v_0\cdot a + v_0\cdot a = 1$

$a = \frac{2}{3v_0}$

(2) 速率大于 v_0 的粒子数 $N_> = \frac{2}{3}\cdot N$

小于 v_0 的粒子数 $N_< = \frac{1}{3}N$

(3) 平均速率

$\bar{v} = \int_0^{v_0} v f(v)dv + \int_{v_0}^{2v_0} v f(v)dv$

$= \int_0^{v_0}\frac{2v^2}{3v_0^2}dv + \int_{v_0}^{2v_0}\frac{2v}{3v_0}dv = \frac{11}{9}v_0$

9.23 $f(v) = 4\pi\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{\frac{-mv^2}{2kT}}$

$= \frac{\sqrt{2}}{\sqrt{\pi}}\left(\frac{m}{kT}\right)^{\frac{3}{2}} v^2 e^{\frac{-mv^2}{2kT}}$

又 $\varepsilon_t = \frac{1}{2}mv^2$　即 $v = \sqrt{\frac{2\varepsilon_t}{m}}$

$f(v)dv = \frac{\sqrt{2}}{\sqrt{\pi}}\left(\frac{1}{kT}\right)^{\frac{3}{2}} m^{\frac{3}{2}} v^2 e^{-\frac{\frac{1}{2}mv^2}{kT}} dv$

$= \frac{\sqrt{2}}{\sqrt{\pi}}(kT)^{-\frac{3}{2}} m^{\frac{3}{2}} v^2 e^{-\frac{\varepsilon_t}{kT}} \frac{2}{m}\frac{1}{2\sqrt{\frac{2\varepsilon_t}{m}}} d\varepsilon_t$

$= \frac{\sqrt{2}}{\sqrt{\pi}}(kT)^{-\frac{3}{2}}\cdot 2\varepsilon_t\cdot e^{-\frac{\varepsilon_t}{kT}}\cdot\frac{1}{\sqrt{2\varepsilon_t}} d\varepsilon_t$

$= \frac{2}{\sqrt{\pi}}(kT)^{-\frac{3}{2}}\cdot \varepsilon_t^{\frac{1}{2}} e^{-\frac{\varepsilon_t}{kT}} d\varepsilon_t$

$= F(\varepsilon_t)d\varepsilon_t$

令 $F'(\varepsilon_t) = \frac{2}{\sqrt{\pi}}(kT)^{-\frac{3}{2}}\left(\frac{1}{2}(\varepsilon_t)^{-\frac{1}{2}} - \frac{\varepsilon_t^{\frac{1}{2}}}{kT}\right)e^{-\frac{\varepsilon_t}{kT}} = 0$

解 $\varepsilon_t = \frac{kT}{2}$　最概然平均动能　　$\frac{1}{2}mv_p^2 = \frac{1}{2}\times m\times\frac{2kT}{m} = kT = 2\varepsilon_t$

9.24 (1) $\ln\frac{n_0}{n} = \frac{mgh}{kT}$

粒子既受重力也受浮力

将 $(\Delta-\delta)\cdot\frac{4}{3}\pi a^3$ 代入 m 可知

$\ln\frac{n_0}{n} = \frac{4\pi a^3(\Delta-\delta)gh}{3kT}$

则 $kT\ln\frac{n_0}{n} = \frac{4}{3}\pi a^3(\Delta-\delta)gh$

又 $\frac{m}{k} = \frac{M}{R}$　可知 $k = \frac{mR}{M} = \frac{R}{N_A}$

故 $\frac{RT}{N_A}\ln\frac{n_0}{n} = \frac{4}{3}\pi a^3(\Delta-\delta)gh$

(2)

$N_A = \frac{RT\ln\frac{n_0}{n}}{\frac{4}{3}\pi a^3(\Delta-\delta)gh}$

$= \frac{8.31\times 293\times\ln\frac{7160}{3360}}{\frac{4}{3}\times 3.14\times(0.212\times10^{-6})^3\times 0.2067\times 9.8\times 30\times10^{6}}$

$= 7.6\times10^{23}\ mol^{-1}$

$\frac{860}{1620}\approx 0.53$

$\frac{1620}{3360}\approx 0.48$

$\frac{3360}{7160}\approx 0.47$

比值近似相等

9.25 (1) $n = \frac{N_A}{V_m} = \frac{6.02\times10^{23}}{3.01\times10^{4}} = 2.00\times10^{19}$

(2) $\Gamma = \frac{1}{4}n\bar{v} = \frac{1}{4}\times 2.00\times10^{19}\times\sqrt{\frac{8\times 8.31\times 373}{\pi\times 18\times10^{-3}}}$

$= 3.31\times10^{23}$

(3) 在饱和状态下，水与水蒸气比例动态平衡

即每秒也有 3.31×10^{23} 个水分子从 $1cm^2$ 面积的水面上跑出

(4) 等温条件下，水蒸气的压强不变，恒为 1 atm

饱和

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9.26 $\left(P_1+\frac{m^2}{M^2}\frac{a}{V^2}\right)\left(V-\frac{m}{M}b\right)=\frac{m}{M}RT$

解 $P_1=\frac{mRT}{MV-mb}-\frac{m^2}{M^2}\frac{a}{V^2}$

$$=\frac{1.1\times8.31\times286}{44\times10^{-3}\times20\times10^{-3}-1.1\times0.0427\times10^{-3}}-\frac{(1.1\times10^{-3})^2\times3.64\times10^5}{(44\times10^{-3})^2\times(20\times10^{-3})^2}$$

$$=2.57\times10^6\ Pa$$

$P_2V=\frac{m}{M}RT$

解 $P_2=\frac{mRT}{MV}=\frac{1.1\times8.31\times286}{40\times10^{-3}\times20\times10^{-3}}=2.97\times10^6\ Pa$

这时CO_2气体内压强 $P=\frac{m^2a}{M^2V^2}=\frac{(1.1\times10^{-3})^2\times3.64\times10^5}{(44\times10^{-3})^2\times(20\times10^{-3})^2}=5.69\times10^5\ Pa$

泻流中速率的平均值

$$\cancel{\bar{v}_x=\frac{\iiint v_x\,\bar{n}(\vec{v})\,dv_x dv_y dv_z}{\iiint \bar{n}(\vec{v})\,dv_x dv_y dv_z}=\frac{\int_0^\infty v_x^2 g(v_x)\,dv_x}{\int_0^\infty v_x g(v_x)\,dv_x}=\left(\frac{\pi kT}{2m}\right)^{\frac{1}{2}}}$$

$\bar{v}_{eff}=\int_0^\infty v f_{eff}(v)\,dv$

又 $f_{eff}=Avf(v)$，$\int_0^\infty f_{eff}(v)\,dv=1$

则 $A\int_0^\infty vf(v)\,dv=1$

即 $A\int_0^\infty 4\pi\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}}v^3e^{-\frac{mv^2}{2kT}}\,dv=1$

令 $B=4\pi\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}}A$ 则 $\int_0^\infty v^3e^{-\frac{mv^2}{2kT}}\,dv=\frac{1}{B}$

令 $u=\frac{mv^2}{2kT}$ 则 $v=\sqrt{\frac{2kTu}{m}}$

$$\left(\frac{2kT}{m}\right)^{\frac{3}{2}}\int_0^\infty u^{\frac{3}{2}}e^{-u}\cdot\frac{\frac{2kT}{m}}{2\sqrt{\frac{2kTu}{m}}}\,du=\frac{1}{B}$$

$$2\left(\frac{kT}{m}\right)^2\int_0^\infty ue^{-u}=\frac{1}{B}$$

$$2\left(\frac{kT}{m}\right)\ -(1+u)e^{-u}\Big|_0^{+\infty}=\frac{1}{4\pi}\cdot\left(\frac{2\pi kT}{m}\right)^{\frac{3}{2}}\cdot\frac{1}{A}$$

$$A=\sqrt{\frac{\pi kT}{8m}}$$

$$\bar{v}_{eff}=\int_0^\infty vf_{eff}(v)\,dv=B\int_0^\infty\left(\frac{2kT}{m}\right)^2u^{\frac{3}{2}}e^{-u}\cdot\sqrt{\frac{kT}{2m}}\,du$$

$$=4\pi\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}}\times\sqrt{\frac{\pi kT}{8m}}\times2\sqrt{2}\times\left(\frac{kT}{m}\right)^{\frac{5}{2}}\int_0^\infty u^{\frac{3}{2}}e^{-u}\,du$$

$$=\sqrt{\frac{2}{\pi}}\times\sqrt{\pi}\times\sqrt{\frac{kT}{m}}\times\frac{3}{4}\sqrt{\pi}$$

$$=3\sqrt{\frac{\pi kT}{8m}}$$

$$=\frac{3}{8}\pi\sqrt{\frac{8kT}{\pi m}}=\frac{3}{8}\bar{v}$$

综上证毕

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